

Final block of the layer-aligned unified transformer ($\ell = 29$)

h_t^A

h_t^L

(a)

MeanPool over $\tilde{h}_t^A$

$\tilde{h}_t^A \in \mathbb{R}^{1024}$

Linear $1024 \rightarrow 512$

SiLU

Linear $512 \rightarrow 1$

predicted progress $\hat{p}_t$



$$\mathcal{L}_{\text{prog}} = \|\hat{p}_t - p_t\|_2^2$$

linear output, no output squashing;
phase target $p_t = k_t / \max(N-1, 1) \in [0, 1]$,
constant within a phase

(b)

$W_A: 1024 \rightarrow 2048$

$W_L: 2048 \rightarrow 2048$

$\text{Mem}_t = [W_A \tilde{h}_t^A; W_L h_t^L]$

Causal self-attention

Cross-attention ($K, V \leftarrow \text{Mem}_t$)

FFN (8192)

lm_head reused from the VLM

$\hat{y}_t$: “pick the blue block”

$$\mathcal{L}_{\text{sub}} = -\sum_j m_{t,j} \log p_{\theta}(y_{t,j} | y_{t, < j}, \text{Mem}_t)$$

$\times 4$ layers, $d = 2048$, 16 heads

embed_tokens
reused from the
VLM

train: teacher
forcing on
 $y_{t, < j}$

test: greedy
decoding, up
to 64 tokens

retained action tokens $\tilde{h}_t^A$

discarded state and register positions

VLM tokens h_t^L

Neither head detaches its input, so both auxiliary losses backpropagate into the unified transformer.
Post-training optimizes $\mathcal{L}_{\text{post}} = \lambda_v \mathcal{L}_{\text{video}} + \lambda_a \mathcal{L}_{\text{action}} + \lambda_s \mathcal{L}_{\text{sub}} + \lambda_p \mathcal{L}_{\text{prog}}$, with $\lambda_v = \lambda_a = 1$ and $\lambda_s = \lambda_p = 0.5$.